Lista - Função do 1º grau - Matemática Elementar

6. Se $f(x) = x^2 + x - 2$, determine $f(2)$, $f(-2)$, $f(a)$, $f(-a)$, $f(a^2)$, $f(a+h)$ e o valor de x tal que $f(x) = 10$.

- $f(2) = 2^2 + 2 - 2$
  $= 4 + \cancel{2} - \cancel{2}$
  $= 4$

- $f(-2) = (-2)^2 + (-2) - 2$
  $= 4 - 2 - 2$
  $= 4 - 4 = 0$

- $f(a) = a^2 + a - 2$
  $= a^2 + a - 2$

- $f(-a) = (-a)^2 + (-a) - 2$
  $= a^2 - a - 2$

- $f(a^2) = (a^2)^2 + a^2 - 2$
  $= a^{2 \cdot 4} + a^2 - 2$
  $= a^8 + a^2 - 2$

- $f(a+h) = (a+h)^2 + (a+h) - 2$
  $\rightarrow a^2 + 2ah + h^2$
  $= a^2 + 2ah + h^2 + a + h - 2$

- $f(x) = x^2 + x - 2 = 10$
  $= x^2 + x - 2 - 12$
  $= x^2 + x - 12$

$a = 1 \quad b = 1 \quad c = -12$

$\Delta = b^2 - 4ac$

$\Delta = 1^2 - 4 \cdot 1 (-12)$

$\Delta = 1 + 48$

$\Delta = 49$

$x = \frac{-b \pm \sqrt{\Delta}}{2a} \rightarrow x = \frac{-1 \pm \sqrt{49}}{2 \cdot 1} \rightarrow x = \frac{-1 \pm 7}{2}$

$\rightarrow x' = \frac{-1+7}{2} = \frac{6}{2} = 3$

$\rightarrow x'' = \frac{-1-7}{2} = \frac{-8}{2} = -4$

7. Se $f(x) = \frac{x^2 - 4}{x - 1}$, achar:

a. $f(0) = \frac{0^2 - 4}{0 - 1} = \frac{0 - 4}{0 - 1} = \frac{-4}{-1} = 4$

b. $f\left(\frac{1}{2}\right) = \frac{\left(\frac{1}{2}\right)^2 - 4}{\left(\frac{1}{2}\right) - 4} = \frac{\left(\frac{1}{4} - \frac{4}{1}\right)}{\left(\frac{1}{2} - \frac{4}{1}\right)} = \frac{\left(\frac{1-16}{4}\right)}{\left(\frac{1-8}{2}\right)} = \frac{-15}{4} = -3,75 = 1,5$

$\frac{-1}{2} \quad -0,5$

$\left(\frac{1^2}{2^2}\right) = \frac{1}{4}$

c. $f(-2) = \frac{(-2)^2 - 4}{-2 - 4} = \frac{4-4}{-3} = \frac{0}{3} = 0$

d. $f(z^2) = \frac{(z^2)^2 - 4}{z^2 - 4} = \frac{z^4 - 4}{z^2 - 4} = \frac{(z^2 - 4)(z^2 + 4)}{z^2 - 4}$

8. Dada a função $(f) = mx + 3$, determine o m valendo que $f(1) = 6$.

$6 = m(1) + 3$ $\qquad f(x) = 3x + 3$

$6 = m + 3$

$m = 6 - 3$

$m = 3$

* 9. A função $f(x)$ é do primeiro grau. Escreva esta função se $f(-1) = 2$ e $f(2) = 3$.

$f(-1) = 2 \rightarrow -a + b = 2$

$f(2) = 3 \rightarrow 2.a + b = 3$

$(2a + b) - (-a + b) = 3 - 2$

$3a = 1$

$a = \frac{1}{3}$

$-a + b = 2$

$-\frac{1}{3} + b = 2$

$b = 2 + \frac{1}{3} = \frac{7}{3}$

$f(x) = \frac{1}{3}x + \frac{7}{3}$

tilibra

10. Obtenha a função do primeiro grau cujo gráfico passa pelos pontos A e B nos seguintes casos:

1a. $A(1,2)$ $\quad m = \frac{3-2}{2-1} = \frac{1}{1} = 1$

$B(2,3)$

Logo $(y-2) = 1(x-1)$

$y-2 = x-1$

$y = x-1+2$

$y = x+1$

Então $f(x) = x+1$

$f(1) = 1+1 = 2$

$f(2) = 2+1 = 3$

Coeficiente Angular

$a = \frac{y1 - y0}{x1 - x0}$

O y sempre vem vezes

$(y - y0) = m(x - x0)$

1b. $A(-1,0)$ $\quad m = \frac{2-0}{4-(-1)} = \frac{2}{4+1} = \frac{2}{5} = 0,4$

$B(4,2)$

Logo $(y-0) = 0,4(x-(-1))$

$y-0 = 0,4x + 0,4$

$y = 0,4x + 0,4$

Então $f(x) = 0,4x + 0,4$

$f(-1) = (0,4(-1)) + 0,4 = -0,4 + 0,4 = 0$

$f(4) = (0,4 \cdot 4) + 0,4 = 1,6 + 0,4 = 2$

1c. $A(2,1)$ $\quad m = \frac{4-1}{0-2} = \frac{3}{-2} = -1,5$

$B(0,4)$

Logo $(y-1) = -1,5(x-2)$

$y-1 = -1,5 \times 3$

$y = -1,5x + 3 + 1$

$y = -1,5x + 4$

Então $f(x) = -1,5x + 4$

$f(2) = -1,5 \cdot 2 + 4 = -3 + 4 = 1$

$f(0) = -1,5 \cdot 0 + 4 = 0 + 4 = 0$

11- Esboce o gráfico e faça o estudo de sinal das funções de 1º grau.

1a- $f(x) = -2x$

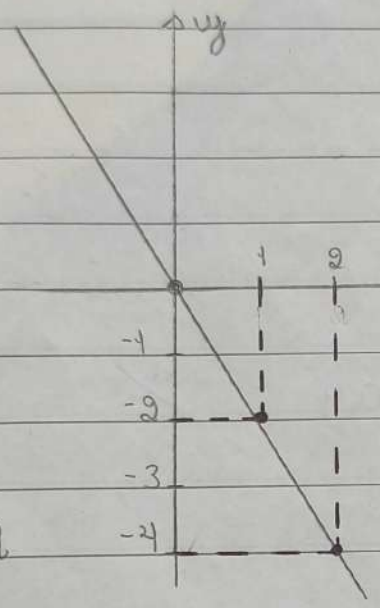
x	f(x)	
0	$-2.0 = 0$	(0,0)
1	$-2.1 = -2$	(1,-2)
2	$-2.2 = -4$	(2,-4)

$f(x) = -2x = 0$

$x = 0$

$x = 2$

$x = 0$



$x = 1 = -2.1 = -2$

$x - 1 = -2 - 1 = 2$

$f(x) < 0$ quando $x < 0$

$f(x) > 0$ quando $x > 0$

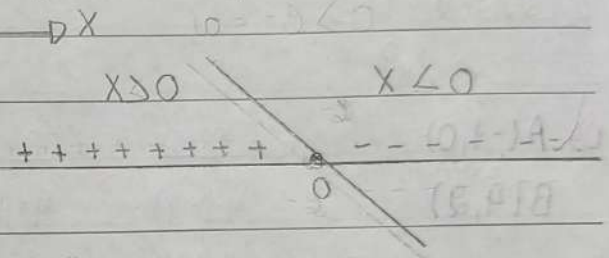


Imagem (f) = ℝ

2b- $f(x) = -2x + 4$

x	f(x)	
0	$-2.0+4 = 4$	(0,4)
1	$-2.1+4 = 2$	(1,2)
2	$-2.2+4 = 0$	(2,0)

$x = 1 = -2.1+4 = -2+4 = -2$

$x = 3 = -3.1+4 = -3+4 = 1$

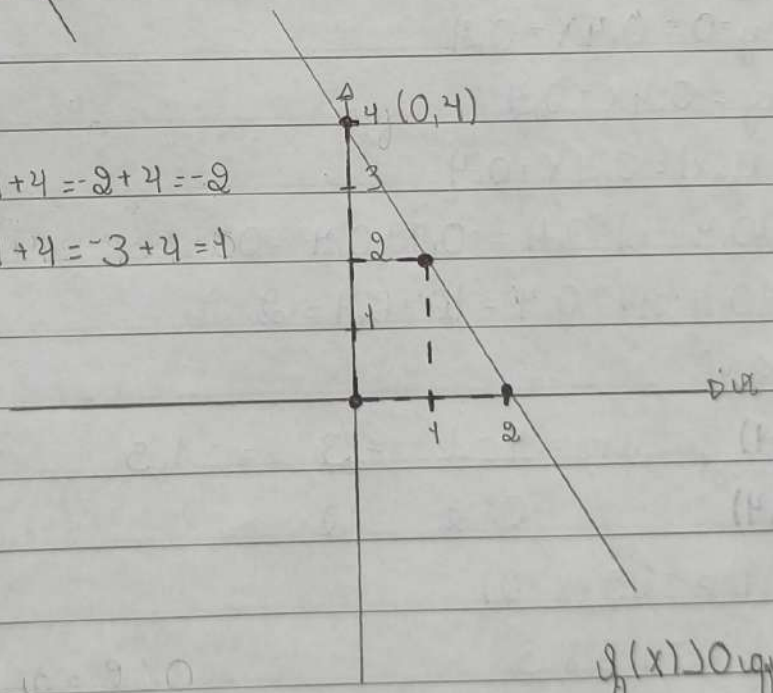


Imagem (f) = ℝ

$f(x) = -2x + 4 = 0$

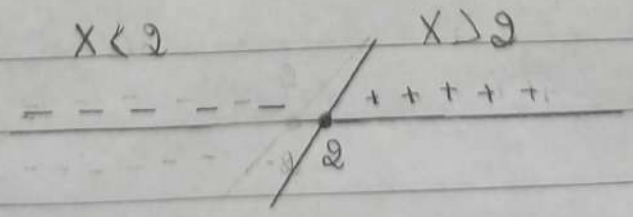
$-2x = -4$

$2x = 4$

$x = \frac{4}{2} = 2$

$f(x) > 0$ quando > 2

$f(x) < 0$ quando < 2



1c. $f(x) = 3x - 1$

X	f(X)
0	3.0-1 = -1 (0,-1)
1	3.1-1 = 2 (1,2)
2	3.2-1 = 5 (2,5)

$x = 1 = 3.1 - 1 = 3 - 1 = 2$

$x = -0,300 = 3.-0,300 - 1$

$= -0,9 - 1$

$= -1,9$

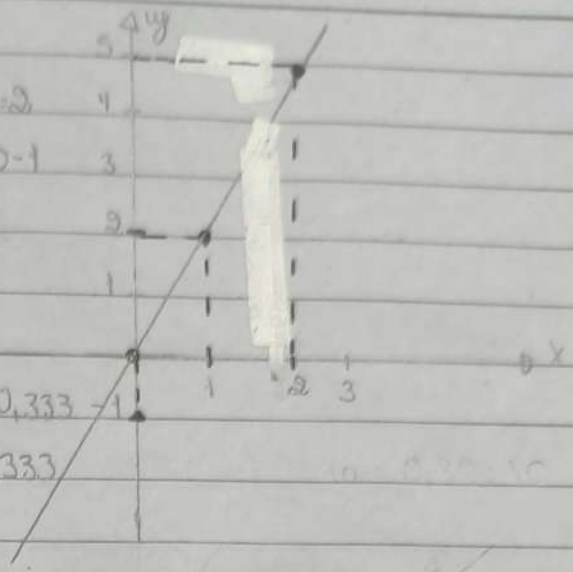


Imagem (f) = IR

$f(x) = 3x - 1$

$x = \frac{-1}{3}$

$x = -0,333$

$f(x) > 0$ quando $> -0,333$

$f(x) < 0$ quando $< -0,333$

$x < -0,333$ $\quad$ $x > -0,333$

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